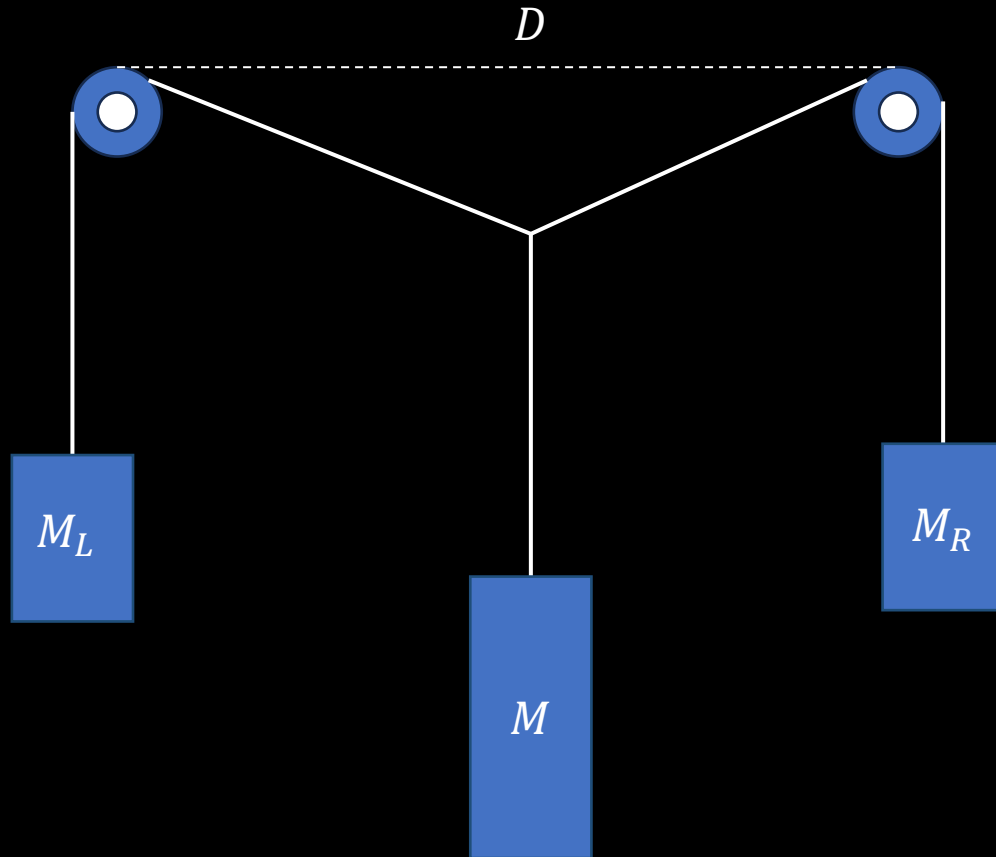
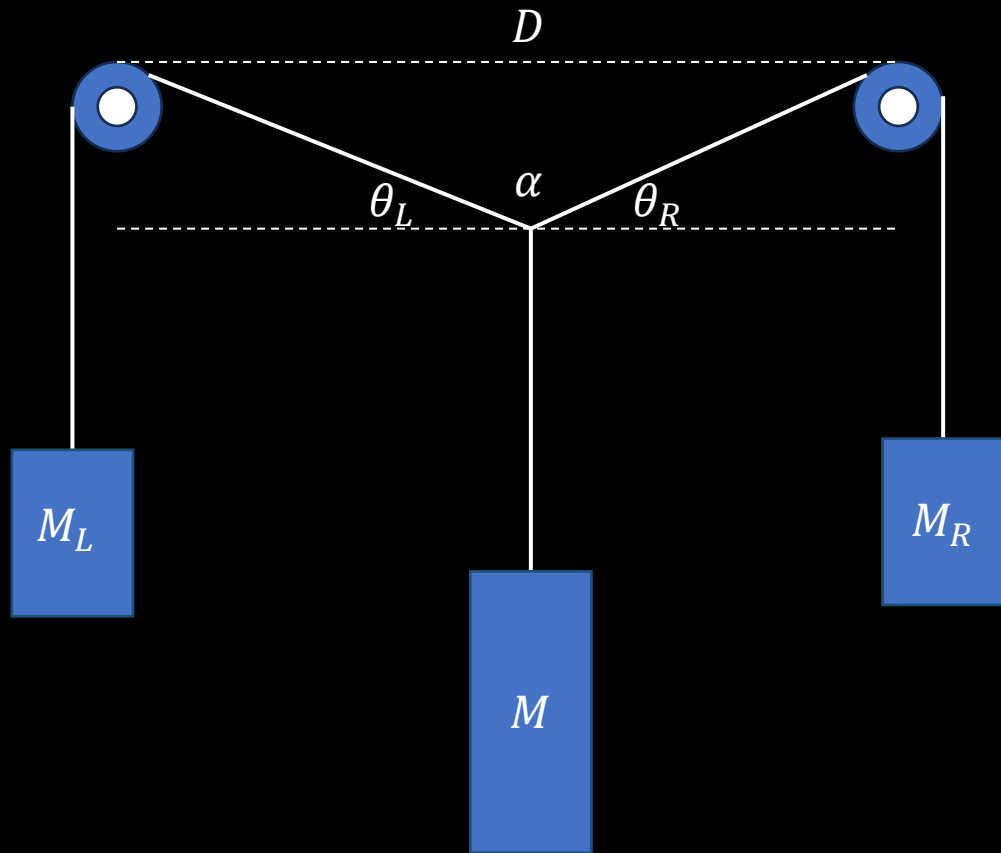
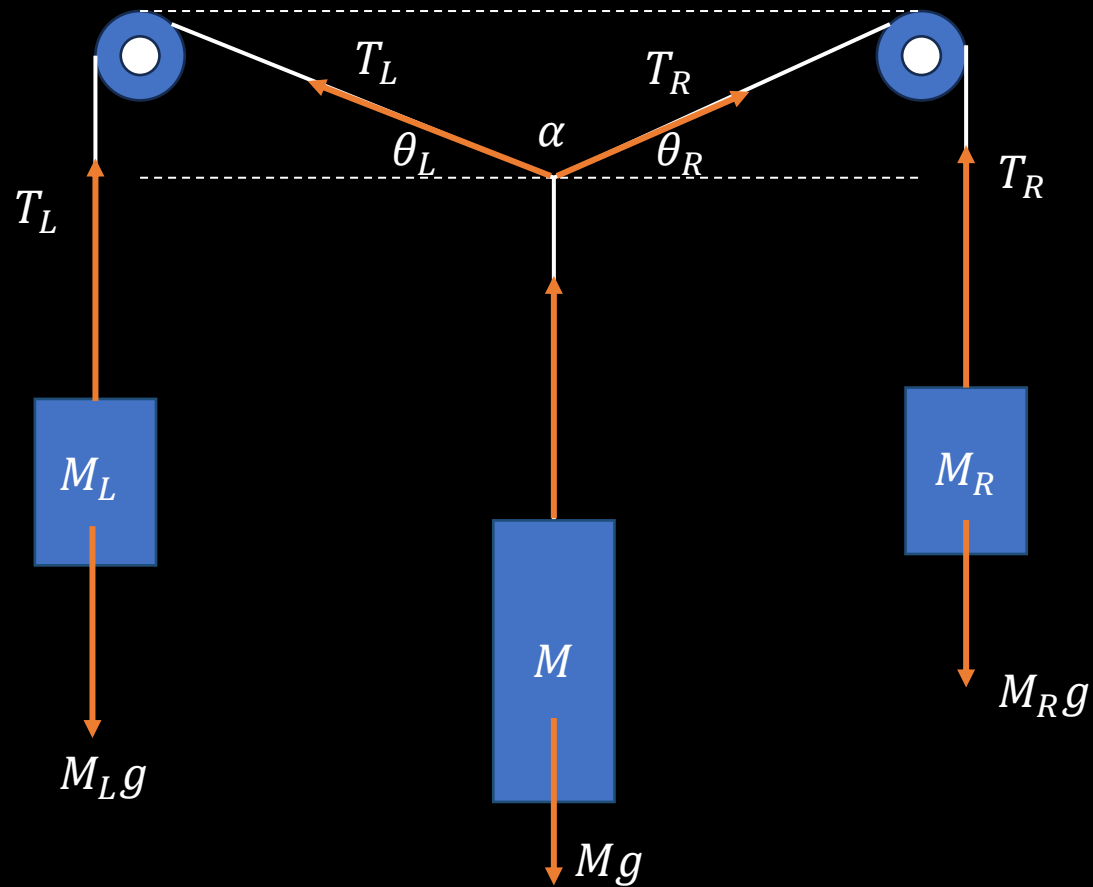


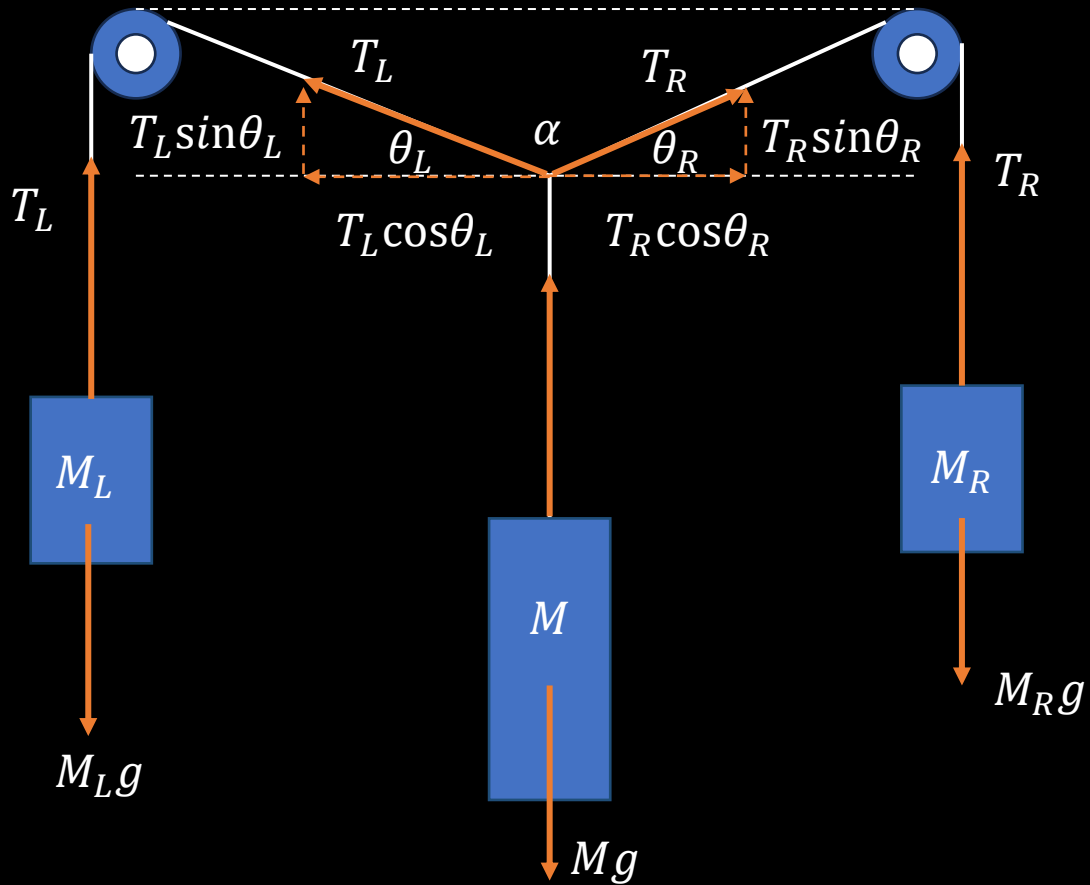
Pitágoras e as polias de Atwood



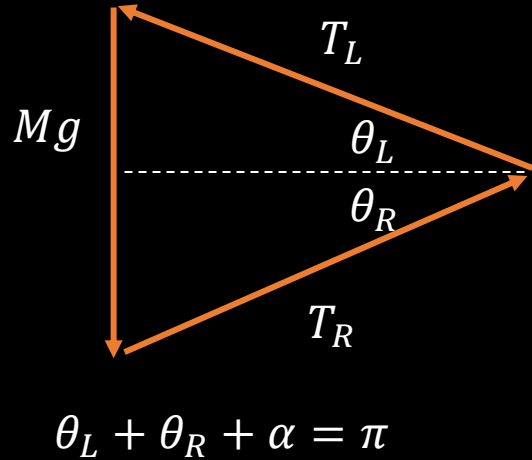


$$\theta_L + \theta_R + \alpha = \pi$$





Regra do paralelogramo



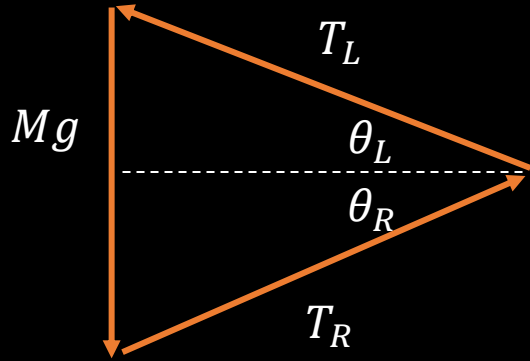
Soma das forças horizontais

$$T_L \cos \theta_L - T_R \cos \theta_R = 0$$

Soma das forças verticais

$$T_L \sin \theta_L + T_R \sin \theta_R = Mg$$

Regra do paralelogramo



$$\theta_L + \theta_R + \alpha = \pi$$

$$M^2 = M_L^2 + M_R^2 - 2M_L M_R \cos(\theta_L + \theta_R)$$

Soma das forças horizontais

$$T_L \cos \theta_L - T_R \cos \theta_R = 0$$

$$\frac{M_L}{M_R} = \frac{\cos \theta_R}{\cos \theta_L}$$

Soma das forças verticais

$$T_L \sin \theta_L + T_R \sin \theta_R = Mg$$

$$M_L \sin \theta_L + M_R \sin \theta_R = M$$

$$\theta_L + \theta_R + \alpha = \pi \tag{01}$$

$$M^2 = M_L^2 + M_R^2 - 2M_L M_R \cos(\theta_L + \theta_R) \tag{02}$$

$$\frac{M_L}{M_R} = \frac{\cos\theta_R}{\cos\theta_L} \tag{03}$$

$$M = M_L \sin\theta_L + M_R \sin\theta_R \tag{04}$$

$$\theta_L + \theta_R + \alpha = \pi \tag{01}$$

$$M^2 = M_L^2 + M_R^2 - 2M_L M_R \cos(\theta_L + \theta_R) \tag{02}$$

$$\theta_L + \theta_R + \alpha = \pi \quad (01)$$

$$\Rightarrow \cos(\theta_L + \theta_R) = \cos(\pi - \alpha)$$

$$M^2 = M_L^2 + M_R^2 - 2M_L M_R \cos(\theta_L + \theta_R) \quad (02)$$

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$$\Rightarrow \cos(\theta_L + \theta_R) = \cos(\pi - \alpha)$$

$$\Rightarrow \cos(\theta_L + \theta_R) = -\cos\alpha$$

$$M^2 = M_L^2 + M_R^2 - 2M_L M_R \cos(\theta_L + \theta_R) \quad (02)$$

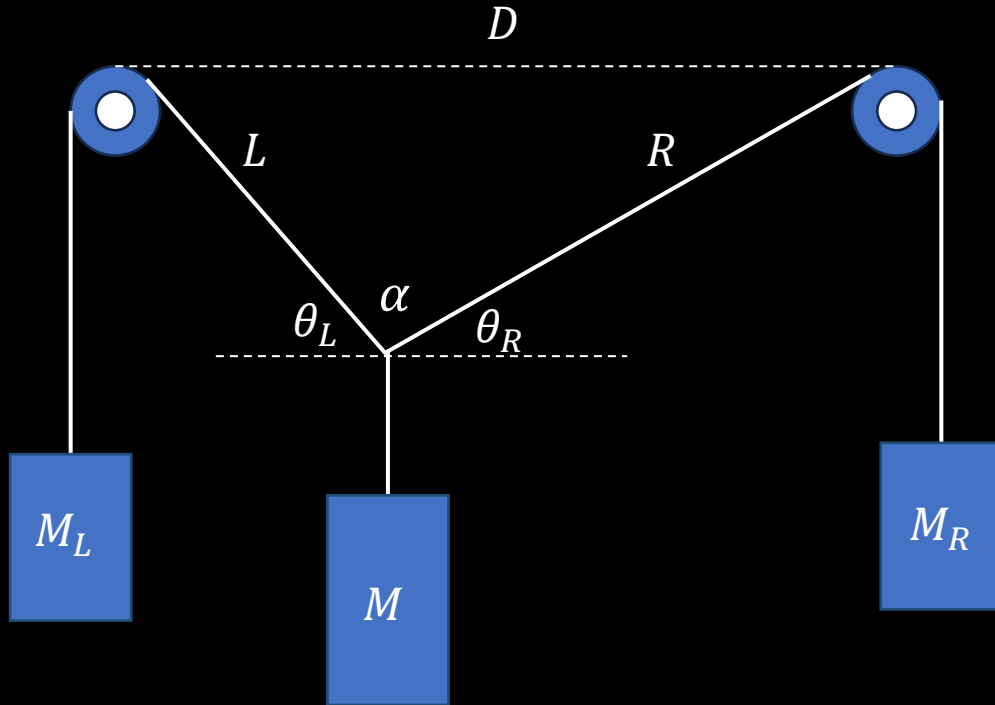
$$\Rightarrow M^2 = M_L^2 + M_R^2 + 2M_L M_R \cos\alpha$$

$$\Rightarrow M^2 = M_L^2 + M_R^2 + 2M_L M_R \cos \alpha$$

$$\Rightarrow \cos \alpha = \frac{M^2 - M_L^2 - M_R^2}{2M_L M_R}$$

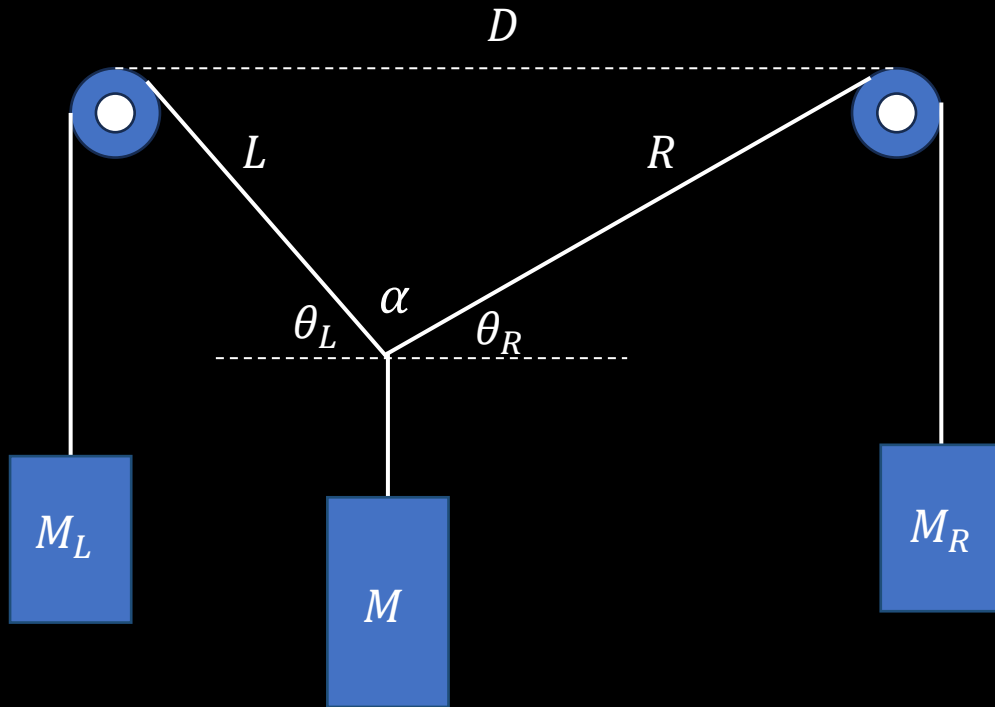
$$\alpha = \frac{\pi}{2} \Rightarrow M^2 = M_L^2 + M_R^2$$

$$\cos\alpha = \frac{M^2 - M_L^2 - M_R^2}{2M_L M_R}$$



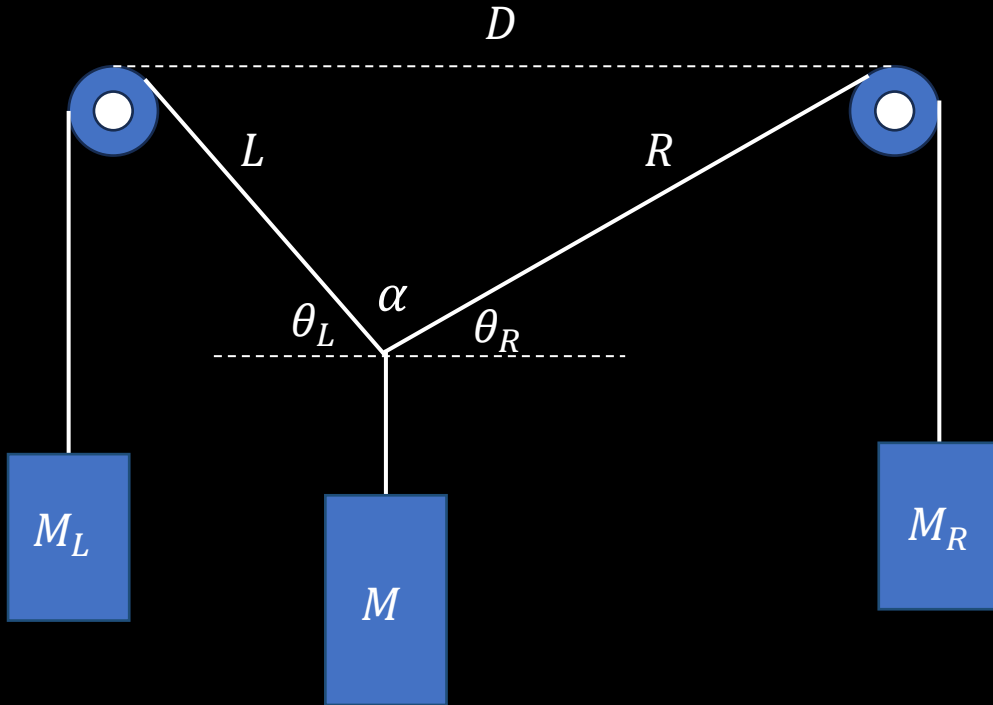
$$\cos\alpha = \frac{M^2 - M_L^2 - M_R^2}{2M_L M_R}$$

$$\cos\alpha = \frac{D^2 - L^2 - R^2}{2LR}$$



$$\cos\alpha = \frac{M^2 - M_L^2 - M_R^2}{2M_L M_R}$$

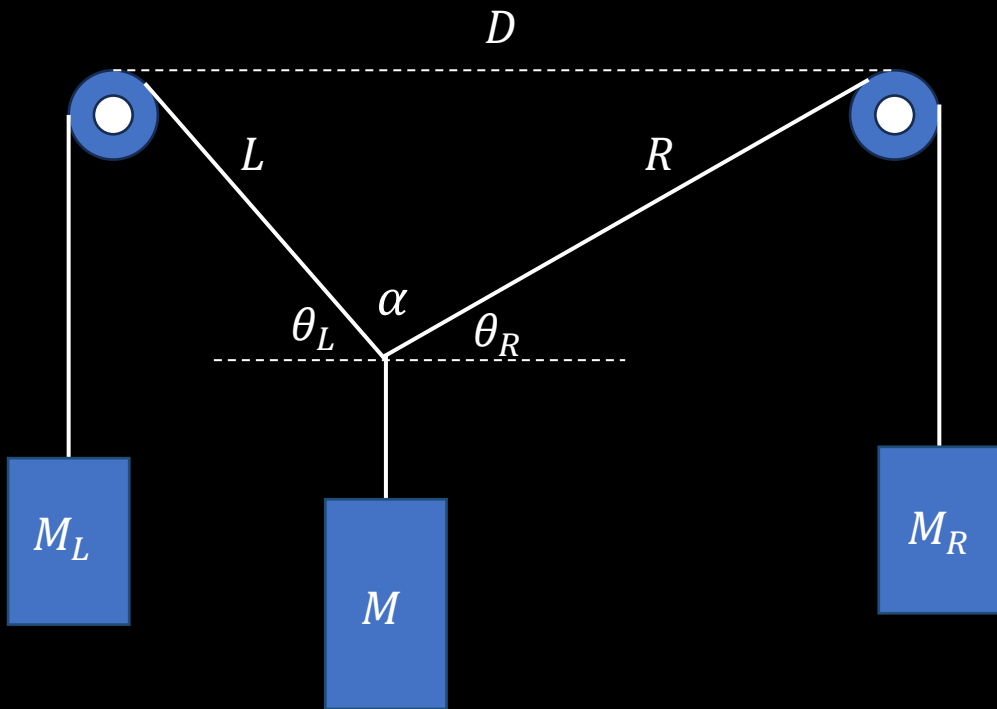
$$\cos\alpha = \frac{D^2 - L^2 - R^2}{2LR}$$



$$\alpha = \frac{\pi}{2} \Rightarrow M^2 = M_L^2 + M_R^2$$

$$\cos\alpha = \frac{M^2 - M_L^2 - M_R^2}{2M_L M_R}$$

$$\cos\alpha = \frac{D^2 - L^2 - R^2}{2LR}$$



$$\alpha = \frac{\pi}{2} \Rightarrow M^2 = M_L^2 + M_R^2$$

$$\alpha = \frac{\pi}{2} \Rightarrow D^2 = L^2 + R^2$$